

Finite Certificates for Surcomplex Wick Summability

Valuation balancing, connected obstructions,
and multiscale Gaussian perturbations

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Abstract

A divergent ordinary perturbation expansion can have an exact surcomplex value, but only when its supports permit a Hahn sum. For a finite polynomial interaction and an arbitrary symmetric covariance matrix over $\mathbb{C}((t^\Gamma))$, we determine exactly when the full family of Wick diagrams is strongly summable. The answer is a finite system of strict valuation inequalities. Equivalently, there is a diagonal monomial change of variables that makes every coupling and every nonzero covariance entry infinitesimal. If this change of variables does not exist, a finite nonnegative integer certificate generates an explicit infinite obstruction to summability.

The same criterion controls every pairable monomial observable. When all interaction vertices have valence at least two, it also controls the family of connected vacuum diagrams: passing to connected diagrams cannot remove a diagramwise Hahn obstruction. For positive-definite real covariance, the criterion reduces to one inequality per interaction monomial. We derive a sharp multiscale stationary-phase condition, exhibit a mixed instability invisible to the individual interactions, and show precisely how complex cancellation can evade any criterion for the ungrouped diagram family.

The results allow arbitrary-rank divisible ordered value groups and arbitrary Hahn-unit tails. They specialize to surreal and surcomplex numbers through Conway normal forms. Full mathematical proofs, an exact rational and lexicographic certificate solver, and 7,684 reproducible finite checks are provided. Classical Wick calculus and support lemmas are explicitly separated from the proposed new finite-certificate theorem package. Everything here is mathematics about finite-dimensional formal Gaussian expansions over a Hahn field; a Hahn value is not an integral, a measure or a physical amplitude, and no claim about physics is made.

Research status. Prepared with AI assistance; not refereed. The Lean ledger covers the support, finite-semigroup and valuation Cauchy–Schwarz lemmas; the main summability and connected-domain theorems remain pending. Proposed novelty is not certified, and no solution of a named longstanding conjecture is claimed. The source’s repository audit is pinned to `4cf691c7d951e037739d32d9f5c387dcce724f3c`; its archive access was incomplete. Section 13 records the source, novelty and verification boundaries, with current-tree corrections and the physics-report comparison. Finite computations are checks, not substitutes for proofs.

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1 The question and the main answer

Gaussian perturbation theory is a useful test of whether surreal scalars can do something mathematically substantive beyond renaming a formal parameter. There is a familiar formal expansion, an enormous supply of diagrams, and a real obstruction to ordinary convergence. The question here is not whether Wick's theorem survives a change of characteristic-zero coefficient field. That part is algebraic and classical. The question is:

For which actual surcomplex couplings and covariance entries does the entire Wick diagram family admit a well-defined, order-independent Hahn sum?

The adjective “actual” matters. A formal series in independent parameters does not automatically admit substitution of specified infinite or infinitesimal numbers. A substitution can make exponents descend forever, or make infinitely many diagrams contribute to one exponent. Conversely, individual parameters need not all be infinitesimal: the incidence relations between vertices and edges can force every realizable product to be small.

Here is the answer in its most usable form. Write the interaction as

$$P(x) = \sum_{a=1}^s g_a x^{\alpha_a}, \quad \lambda_a = v(g_a), \quad c_{ij} = v(C_{ij}),$$

where C is a symmetric covariance matrix, zero entries are omitted, and the α^a are nonzero vectors of nonnegative integers. Let Γ be a nontrivial divisible ordered abelian group. Then the diagram family is strongly Hahn summable if and only if there are $p_1, \dots, p_d \in \Gamma$ with

$$\lambda_a + \sum_i \alpha_i^a p_i > 0 \quad (1 \leq a \leq s), \quad (1.1)$$

$$c_{ij} - p_i - p_j > 0 \quad (C_{ij} \neq 0). \quad (1.2)$$

For a loop, the second expression is $c_{ii} - 2p_i$. Thus the answer is a *finite strict feasibility problem*, even though the family being summed is infinite.

The inequalities have a direct meaning. Under $x_i = t^{p_i} y_i$, the new couplings and covariances are

$$\tilde{g}_a = g_a t^{\alpha_a \cdot p}, \quad \tilde{C}_{ij} = C_{ij} t^{-p_i - p_j}.$$

All their valuations become positive. The two changes compensate exactly along a Wick contraction. In vacuum diagrams the compensation is complete; with an external monomial x^β , the only remaining factor is $t^{p \cdot \beta}$. This converts a potentially mixed-scale problem into a legitimate substitution into a formal series in finitely many infinitesimals.

1.1 What is substantial about the criterion?

Necessity is as important as sufficiency. A failed balancing problem yields nonnegative integers m_a and k_{ij} satisfying the exact color-balance relations

$$\sum_a m_a \alpha_i^a = 2k_{ii} + \sum_{j \neq i} k_{\min(i,j), \max(i,j)}$$

and the nonpositive weight condition

$$\sum_a m_a \lambda_a + \sum_{i \leq j} k_{ij} c_{ij} \leq 0.$$

Repeated copies of this finite configuration force a descending sequence of leading exponents or infinitely many contributions at a single exponent. There is no hidden appeal to numerical convergence or to a rank-one norm.

The finite criterion also has two nontrivial consequences. First, when vertices have valence at least two, an obstructing indecomposable configuration can be rewired into connected configurations of every multiple size. The connected expansion has exactly the same diagramwise domain. Second, positive-definite real covariance eliminates mixed instabilities: the whole criterion collapses to

$$\lambda_a + \frac{1}{2} \sum_i \alpha_i^a c_{ii} > 0 \quad \text{for every } a.$$

Neither assertion follows just by writing down Wick's formula over a bigger field. They concern the admissibility of its infinite specialization.

1.2 Classical ingredients and the proposed contribution

Conway normal forms and the associated notion of summability are standard surreal foundations; Berarducci–Mantova provide a modern account [2, Sections 2.3–2.6]. Neumann's ordered-series lemma supplies the positive-support substitution principle [1]. Classical Wick pairing formulas and the connected-diagram logarithm are described, for example, in Etingof's treatment [3, Chapter 3]. We rederive the finite combinatorics needed here rather than present them as new.

The proposed contribution is the joint theorem consisting of the exact finite-certificate characterization, arbitrary-rank strict balancing, observable-sector equivalence, connected obstruction amplification, and the sharp positive-covariance and mixed-scale consequences. The proofs are self-contained apart from the standard ordered-series support lemma and the standard surreal normal-form interpretation. The supporting finite semigroup and linear-inequality facts are proved explicitly, while remaining classical tools rather than priority claims.

There is an essential restriction: *diagramwise* summability means that the individual diagram terms may be summed without cancellation-dependent infinite regrouping. An isotropic example in Section 11 has a perfectly finite order-grouped answer while its diagram family is not summable. Any theorem omitting that distinction would be false.

1.3 Words and letters with one meaning in this report

Several words used here carry other meanings in other reports of the collection to which this report belongs. Within this report each has exactly the following meaning.

- *Gaussian* refers to the finite formal Gaussian expectation E_C of Section 3 and to the perturbation expansions and models built from it. No measure or integral is meant; the two places that mention a Gaussian integral or function (Sections 8 and 10) do so only to say that none is constructed. The word has nothing to do with Gaussian rationals, Gaussian elimination, Gaussian quadrature, or the Gaussian regulator of the physics report, all of which occur elsewhere in the collection.
- A *moment* is a Wick moment $E_C[x^M]$, a finite sum over pairings (Lemma 3.1). Power moments of measures, Fourier moments and Prony input data are not meant.
- *Positive*, for an element of $\mathbb{K}_{\mathbb{R}}$, is ordered-field positivity: a positive leading coefficient. *Coefficientwise nonnegative*, the condition of Section 11, asks every coefficient to be a nonnegative

real number. For a nonzero element it implies positivity but not conversely, and the two are never interchanged.

- A *Hilbert basis* is the finite indecomposable generating set of the integer semigroup S (Proposition 4.2); no Hilbert space occurs.
- The *Wick valuation cone* is the finite list of strict inequalities described after Corollary 6.3. A datum is *admissible* when it satisfies the equivalent conditions of Theorem 6.1, that is, when its Wick diagram family is diagramwise strongly summable. Other reports use “cone” and “admissible” in other senses.
- A *phase* is the argument of a nonzero complex coefficient, and *stationary phase* is the name of the formal expansion treated in Section 8. Neither refers to the phase symbols fixed in the differential-equations report and adopted by the physics report.
- D is the incidence matrix of Section 3, never a derivation. The symbol ∂_j in Section 9 is formal differentiation in the polynomial variable x_j , as stated after Theorem 9.2.

A few letters are reused locally, each time inside one delimited passage. The reuses most likely to mislead are these. The letter h denotes an element of S (usually of the Hilbert basis), except that in Corollary 8.3 and (8.7) it is $v(\hbar)$ and in Section 10.2 and the higher-rank example of Section 10.4 it is a coupling of the cross-cubic model. The letter a indexes interaction types, except for the exposed slots a_r, b_r in Lemma 7.2 and for $a_i = v(q_i)$ in Corollary 8.3. The letter s counts interaction monomials, except for the cross-edge parameter of the mixed quartic model in Section 10.3. The letter u denotes positive-valuation substitution inputs in Corollary 2.3 and the scaled couplings of (6.2), but the action coefficients in (8.5). The letter q denotes an integer count vector, except for the rational coordinates in the embedding examples after Proposition 2.4 and the quadratic coefficients q_i in (8.5).

2 Hahn fields, strong sums, and surcomplex realization

2.1 The fixed workspace

Throughout, $\mathbb{N}$ includes zero. Let $\Gamma \neq \{0\}$ be a set-sized divisible ordered abelian group. Divisibility means that multiplication by every positive integer is bijective; hence Γ is an ordered $\mathbb{Q}$ -vector space. No Archimedean or finite-rank hypothesis is imposed. Put

$$\mathbb{K} = \mathbb{C}((t^\Gamma)), \quad \mathbb{K}_{\mathbb{R}} = \mathbb{R}((t^\Gamma)).$$

An element of $\mathbb{K}$ is a formal expression

$$f = \sum_{\gamma \in S} a_\gamma t^\gamma$$

with complex coefficients and well-ordered support $S \subseteq \Gamma$ in the *increasing* order. For $f \neq 0$, define

$$v(f) = \min \text{supp}(f), \quad v(0) = +\infty.$$

The symbol $+\infty$ is adjoined for valuation comparisons, not a field element. The valuation satisfies $v(fg) = v(f) + v(g)$ and $v(f + g) \geq \min(v(f), v(g))$. Nonzero complex constants have valuation zero. In particular, factorial growth of complex coefficients does not change valuations.

The real subfield is ordered by the sign of the leading coefficient. An element of positive valuation is infinitesimal relative to every positive real constant. This use of “infinitesimal” concerns the chosen Hahn valuation; we do not use an unspecified topology on a proper class.

Definition 2.1 (Strong summability). A set-indexed family $(f_j)_{j \in J}$ in $\mathbb{K}$ is *strongly summable* if $\bigcup_j \text{supp}(f_j)$ is well ordered and, for each $\gamma \in \Gamma$, only finitely many j satisfy $\gamma \in \text{supp}(f_j)$. Its sum is obtained by adding those finitely many coefficients at each exponent.

A subfamily of a strongly summable family is strongly summable. Finite regrouping of its terms preserves its sum. Multiplication by a fixed Hahn series preserves strong summability; this follows from the same support lemma that defines Hahn multiplication. The converse for arbitrary finite blocks does not hold: cancellation inside infinitely many different blocks can erase an otherwise illegal family. We will use that distinction rather than assume it away.

2.2 The support lemma and substitution

Lemma 2.2 (Neumann’s positive-support lemma, classical). *If $B \subseteq \Gamma_{>0}$ is well ordered, the additive monoid generated by B is well ordered. Each exponent has only finitely many representations as a finite multiset of members of B .*

This is the ordered-series support principle from [1]. It applies in arbitrary ordered abelian groups: it does not require $n\eta \rightarrow +\infty$ for a positive η . We use it in the following standard form, whose precise summability assertion is worth recording.

Corollary 2.3 (Finite infinitesimal substitution). *Let $u_1, \dots, u_r \in \mathbb{K}$ have positive valuations. For every family of complex numbers $(a_n)_{n \in \mathbb{N}^r}$, the family*

$$(a_n u_1^{n_1} \cdots u_r^{n_r})_{n \in \mathbb{N}^r}$$

is strongly summable. Consequently there is an evaluation homomorphism $\mathbb{C}[[U_1, \dots, U_r]] \rightarrow \mathbb{K}$ taking U_j to u_j ; it respects the coefficientwise formal identities involved in addition and multiplication.

Proof. The union $B = \bigcup_j \text{supp}(u_j)$ is a finite union of well-ordered sets and is contained in $\Gamma_{>0}$, so it is well ordered. Every exponent occurring in one of the displayed terms is a sum of members of B . By Lemma 2.2, the union of these supports is well ordered. A fixed exponent has finitely many multiset decompositions in B . Each such finite multiset has only finitely many ways to label its entries by the r series from which they came. Thus only finitely many multiindices n can contribute to that exponent. This proves strong summability.

The identical decomposition argument shows that products of two evaluated series agree with the evaluation of their Cauchy product. Every coefficient comparison involves only a finite sum. Constants and addition are immediate. No bound on $|a_n|$ was used. $\square$

The conclusion is stronger than evaluating convergent complex power series: the formal series may have ordinary radius of convergence zero. It is also different from Borel–Laplace summation, which is an analytic procedure with additional growth and continuation requirements; see [5] for that separate theory. Here the input is a support family and the output is a Hahn series, not a holomorphic function on a sector.

2.3 Embedding the answer into the surcomplex field

Proposition 2.4 (Surcomplex realization). *Suppose an order-preserving additive embedding $\iota : \Gamma \hookrightarrow (\mathbf{No}, +)$ has been fixed. Then*

$$\begin{aligned} \Phi : \mathbb{K} &\longrightarrow \mathbf{No}[i], \\ \sum_{\gamma} (a_{\gamma} + ib_{\gamma})t^{\gamma} &\longmapsto \sum_{\gamma} a_{\gamma}\omega^{-\iota(\gamma)} + i \sum_{\gamma} b_{\gamma}\omega^{-\iota(\gamma)} \end{aligned}$$

is an injective field homomorphism. It carries strongly summable families to their normal-form sums.

Proof. An increasing well-ordered set of γ becomes a reverse well-ordered set of exponents $-\iota(\gamma)$ in the normal-form convention. Both real series therefore define surreal numbers. Uniqueness of normal form gives injectivity. Additivity and the convolution product follow from the normal-form addition and multiplication laws, since ι is additive. The two finiteness conditions for a strong sum are preserved by the injective map on exponents. These are precisely the normal-form operations described in [2, Sections 2.3–2.6]. $\square$

Unless another group is specified, examples with scalar rational exponents are taken in $\Gamma = \mathbb{Q}$, with $t = t^1$ and $\iota(q) = q$. Then t becomes ω^{-1} . They also embed in any nontrivial divisible ordered group after choosing a positive γ_0 and replacing each exponent q by $q\gamma_0$; there is no canonical unit exponent in an arbitrary Γ . For $\Gamma = \mathbb{Q}^r$ in lexicographic order, an explicit choice is

$$\iota(q_1, \dots, q_r) = \sum_{j=1}^r q_j \omega^{r-j}.$$

In rank two, $t^{(0,1)}$ becomes ω^{-1} while $t^{(1,0)}$ becomes $\omega^{-\omega}$. Thus the later higher-rank examples describe genuinely separated surreal scales, not merely a different notation for real powers. All families used below are set-indexed, and all their values live in one fixed set-sized Hahn workspace before this embedding is applied.

3 The Wick family as an integer incidence problem

3.1 Data and conventions

Fix $d, s \geq 1$ and a polynomial

$$P(x) = \sum_{a=1}^s g_a x^{\alpha^a}, \quad g_a \in \mathbb{K}^{\times}, \quad \alpha^a \in \mathbb{N}^d \setminus \{0\}. \quad (3.1)$$

Combine duplicate monomials and remove zero coefficients before applying the results. Let $C = (C_{ij})$ be a symmetric matrix over $\mathbb{K}$, and let

$$E = \{(i, j) : 1 \leq i \leq j \leq d, C_{ij} \neq 0\}.$$

The matrix can be singular. Except in Section 8, it need not be real or positive. It is a *symmetric bilinear* covariance, not a Hermitian covariance: no complex conjugation appears in a Wick contraction.

Let A be the $d \times s$ matrix with columns α^a . Define the $d \times |E|$ matrix D by assigning to an edge $e = (i, j)$ the column

$$D_e = \begin{cases} 2\mathbf{e}_i, & i = j, \\ \mathbf{e}_i + \mathbf{e}_j, & i < j. \end{cases}$$

We write $m = (m_a) \in \mathbb{N}^s$ for the number of vertices of each type and $k = (k_e) \in \mathbb{N}^E$ for the number of contractions of each color type. Edges with zero covariance are not included.

For an external monomial x^β , where $\beta \in \mathbb{N}^d$, define

$$Q_\beta = \{(m, k) \in \mathbb{N}^s \times \mathbb{N}^E : Am + \beta = Dk\}, \quad S = Q_0. \quad (3.2)$$

An observable is called *pairable* when $Q_\beta \neq \emptyset$. The empty vacuum diagram corresponds to $(m, k) = (0, 0) \in S$.

3.2 Counting the contractions

For a vector of nonnegative integers M , let $\mathbf{E}_C[x^M]$ be the sum over all pairings of $|M|$ distinguishable slots, of which M_i have color i , with a pair of colors i, j contributing C_{ij} . The empty moment is 1; odd moments vanish. This defines a linear map from $\mathbb{K}[x_1, \dots, x_d]$ to $\mathbb{K}$. No integration or measure is needed.

Lemma 3.1 (Colored matching multiplicity). *If $Dk = M$, the number of pairings with edge counts k is*

$$W(M, k) = \frac{\prod_i M_i!}{\prod_{(i,i) \in E} 2^{k_{ii}} k_{ii}! \prod_{\substack{(i,j) \in E \\ i < j}} k_{ij}!}. \quad (3.3)$$

In particular, this is a positive integer. The moment is the finite sum $\mathbf{E}_C[x^M] = \sum_{Dk=M} W(M, k) C^k$.

Proof. At color i , partition the M_i distinguishable slots into $2k_{ii}$ slots for loops and k_{ij} slots for every neighboring color j . This contributes

$$\frac{M_i!}{(2k_{ii})! \prod_{j \neq i} k_{ij}!}.$$

Pair the loop slots in $(2k_{ii})!/(2^{k_{ii}} k_{ii}!)$ ways. For $i < j$, match the two sets of k_{ij} slots by a bijection in $k_{ij}!$ ways. Multiplication cancels one of the two cross-edge factorials and gives (3.3). When a loop is not allowed, its count is zero and its factor is 1. There are only finitely many feasible k because the total number of slots is fixed. $\square$

The formal expansion of $\mathbf{E}_C[x^\beta e^{P(x)}]$ suggests the following precise family, which we take as the definition before any sum is asserted.

Definition 3.2 (Wick atoms). For $(m, k) \in Q_\beta$, put

$$T_\beta(m, k) = \frac{W(Am + \beta, k)}{\prod_a m_a!} g^m C^k. \quad (3.4)$$

The *diagramwise Wick sum* $\mathcal{Z}_\beta$ exists when this family is strongly summable; in that case $\mathcal{Z}_\beta = \sum_{Q_\beta} T_\beta(m, k)$. Write $Z = \mathcal{Z}_0$. If Q_β is empty, define $\mathcal{Z}_\beta = 0$.

Each atom collects only finitely many pairings, all with the same monomial $g^m C^k$ and positive rational multipliers. Thus the atom family is strongly summable exactly when the individually indexed pairing family is strongly summable. Their supports agree within every such finite block; there is no cancellation in passing from pairings to (3.4).

Set

$$\lambda_a = v(g_a), \quad c_e = v(C_e), \quad L(m, k) = \sum_a m_a \lambda_a + \sum_e k_e c_e. \quad (3.5)$$

The coefficient in (3.4) is nonzero in characteristic zero, so

$$v(T_\beta(m, k)) = L(m, k). \quad (3.6)$$

This equality is exact, even when the couplings and covariances have arbitrary higher Hahn terms. It is the source of the sharp necessity statements.

4 Finite semigroups and observable sectors

The infinite diagram problem already contains finite combinatorics. We give the elementary arguments to avoid relying on a black-box finite-generation claim.

Lemma 4.1 (Dickson's finiteness lemma, classical). *Every subset of $\mathbb{N}^r$ has finitely many minimal elements for the coordinatewise order. Every element lies above one of these minimal elements.*

Proof. Every infinite sequence in $\mathbb{N}$ has an infinite nondecreasing subsequence: either a value occurs infinitely often, or the sequence is unbounded on every sufficiently chosen tail and one chooses increasing values. Applying this observation successively to each of the r coordinates produces, from any infinite sequence in $\mathbb{N}^r$, an infinite coordinatewise nondecreasing subsequence. Therefore an infinite antichain cannot exist. Distinct minimal elements form an antichain, proving finiteness. Finally, descending strictly below a specified point decreases the sum of its coordinates, so descent must terminate at a minimal element of the given subset. $\square$

Proposition 4.2 (A finite Hilbert basis). *The coordinatewise minimal elements of $S \setminus \{0\}$ form a finite set $\mathcal{H}$. They are exactly the nonzero elements that cannot be written as a sum of two nonzero elements of S . Every element of S is a nonnegative integer combination of elements of $\mathcal{H}$.*

Proof. If $0 \neq h' \leq h$ lie in S , then $h - h' \in S$ because $Am = Dk$ is a homogeneous linear equation. Unless $h = h'$, this is a nontrivial decomposition. Conversely, a decomposition $h = h' + h''$ with both terms nonzero produces a strictly smaller nonzero element $h' \leq h$. This proves the characterization. Finiteness follows from Lemma 4.1. If an element is not minimal, subtract a smaller nonzero element and continue; induction on the sum of all coordinates terminates in elements of $\mathcal{H}$. $\square$

We use ‘‘Hilbert basis’’ for this unique indecomposable generating set of the nonnegative integer kernel. It is a structural finite object; no efficient algorithm for finding it is assumed.

Proposition 4.3 (Finite generation of every observable sector). *If $Q_\beta \neq \emptyset$, let $\mathcal{P}_\beta$ be its finite set of coordinatewise minimal elements. Then*

$$Q_\beta = \bigcup_{q \in \mathcal{P}_\beta} (q + S). \quad (4.1)$$

The union need not be disjoint.

Proof. Choose a minimal $q' \in Q_\beta$ below $q \in Q_\beta$. Subtracting the two inhomogeneous equations gives $A(m - m') = D(k - k')$, with all differences nonnegative. Hence $q - q' \in S$. Conversely, addition of any element of S to an element of Q_β preserves its defining equation. $\square$

In particular, a nonempty sector Q_β is finite if and only if $S = \{0\}$. If $S = \{0\}$, (4.1) says $Q_\beta = \mathcal{P}_\beta$, which is finite. Conversely, if $q_0 \in Q_\beta$ and $0 \neq h \in S$, then the distinct vectors $q_0 + nh$, $n \in \mathbb{N}$, all belong to the sector. This is a statement about the number of atoms, not about the possibly infinite support of an individual Hahn atom.

The same vacuum semigroup therefore controls every external polynomial sector. Proposition 4.3 alone does not establish summability, because translated monomial families can still have bad supports; it identifies the finite repetition mechanism that the valuation criterion will control.

5 A strict alternative over arbitrary ordered value groups

We require a strict linear-inequality theorem over Γ , not merely over $\mathbb{R}$. It is a version of the classical theorem of alternatives, with a short elimination proof that records exactly what divisibility supplies.

Theorem 5.1 (Strict rational alternative). *Let $M \in \mathbb{Q}^{N \times d}$ and $b \in \Gamma^N$. Exactly one of the following possibilities occurs:*

- (i) *there is $p \in \Gamma^d$ such that $Mp + b > 0$ in every coordinate;*
- (ii) *there is $0 \neq r \in \mathbb{Q}_{\geq 0}^N$ such that $r^T M = 0$ and $r \cdot b \leq 0$.*

In particular,

$$\exists p (Mp + b > 0) \iff r \cdot b > 0 \text{ for every } 0 \neq r \in \mathbb{Q}_{\geq 0}^N \text{ with } r^T M = 0. \quad (5.1)$$

Proof. The two possibilities cannot coexist: a nonzero nonnegative rational combination of strictly positive elements of an ordered $\mathbb{Q}$ -vector space is strictly positive.

For existence, eliminate the last variable from a finite system of strict inequalities. Normalize every row with positive last coefficient to

$$p_d + u(p_1, \dots, p_{d-1}) > 0$$

and every row with negative last coefficient to

$$-p_d + w(p_1, \dots, p_{d-1}) > 0.$$

Rows with zero last coefficient are retained. The necessary and sufficient conditions for a last coordinate to exist are the retained rows together with

$$u + w > 0 \quad (5.2)$$

for every lower-bound/upper-bound pair. Necessity is addition. For sufficiency, after fixing the first $d - 1$ coordinates, the finitely many lower bounds $-u$ have a maximum and the finitely many upper bounds w have a minimum. All pairwise inequalities (5.2) say that the former is strictly below the latter. Their midpoint is available by divisibility. If there is only a lower bound, add any positive element of Γ to its maximum; if there is only an upper bound, subtract one from its minimum. If neither occurs, choose zero.

Repeat this elimination until no variables remain. Every generated row is a nonzero nonnegative rational combination of original rows: normalization uses a positive rational multiplier, and new rows are sums of two such rows. Carry the coefficient vector of this combination as part of each row. If a final constant row is nonpositive, its carried vector is the required r : all variable coefficients have been eliminated. A constant row with zero variable coefficients can equivalently be detected at any earlier stage. If every final constant row is positive, recover variables in reverse order by the midpoint or one-sided choices just described. This constructs p and exhausts the alternatives. $\square$

No separation theorem over a real normed space was used. In particular, the proof remains valid for lexicographic groups and for groups of infinite rank. There is no need to choose a real-valued functional that forgets smaller scales.

For the Wick problem, take the rows of M to be

$$M = \begin{pmatrix} A^T \\ -D^T \end{pmatrix}, \quad b = (\lambda, c). \quad (5.3)$$

A nonnegative rational kernel vector $r = (m, k)$ satisfies $Am = Dk$. Multiplication by a common positive denominator makes it an element of S . Consequently Theorem 5.1 gives the exact bridge between valuation balancing and positivity on all realizable diagram counts.

Remark 5.2 (Finite certificate, not a complexity claim). Fourier–Motzkin elimination can generate exponentially many rows. The importance here is that it produces an independently checkable finite witness. A successful output is a vector p satisfying the original strict inequalities. An unsuccessful output is an integer incidence vector $q \in S \setminus \{0\}$ with $L(q) \leq 0$. Either output can be verified by finite rational operations and order comparisons. The supplied code imposes an explicit row budget rather than disguising a resource failure as a mathematical obstruction.

6 The finite-certificate summability theorem

Theorem 6.1 (Exact diagramwise Hahn domain). *For the finite interaction and covariance data of Section 3, the following are equivalent:*

- (i) *The vacuum Wick atom family is strongly summable.*
- (ii) *$L(q) > 0$ for every $q \in S \setminus \{0\}$.*
- (iii) *$L(h) > 0$ for every element h of the finite Hilbert basis $\mathcal{H}$.*
- (iv) *There is $p \in \Gamma^d$ such that*

$$\lambda_a + \alpha^a \cdot p > 0 \quad (1 \leq a \leq s), \quad c_{ij} - p_i - p_j > 0 \quad ((i, j) \in E). \quad (6.1)$$

- (v) *For every $\beta \in \mathbb{N}^d$, the Wick atom family in Q_β is strongly summable.*

Moreover, for any one fixed β with $Q_\beta \neq \emptyset$, strong summability in that single sector is equivalent to all these conditions. The result allows arbitrary Hahn-unit tails and does not depend on the sizes or phases of the nonzero leading complex coefficients.

Proof. (i) implies (ii). Suppose $q \neq 0$ belongs to S and $L(q) \leq 0$. For every $n \geq 1$, the vector nq is a distinct member of S . By (3.6), its atom has leading exponent $nL(q)$. If $L(q) < 0$, these

exponents form a strictly descending sequence in the union of supports. If $L(q) = 0$, exponent zero belongs to the supports of infinitely many atoms. Each case violates Definition 2.1.

(ii) is equivalent to (iii). Necessity is immediate. Conversely, every nonzero element of S is a sum of at least one element of $\mathcal{H}$ by Proposition 4.2. Its weight is a sum of strictly positive elements.

(ii) is equivalent to (iv). Apply Theorem 5.1 to (5.3). Clearing denominators identifies every possible nonnegative rational obstruction with a nonzero element of S of nonpositive weight. Conversely, an integer element of S is already an allowed rational obstruction. This proves the equivalence.

(iv) implies (v). Define

$$u_a = g_a t^{\alpha_a \cdot p}, \quad w_{ij} = C_{ij} t^{-p_i - p_j}. \quad (6.2)$$

All these finitely many Hahn elements have positive valuation. For $(m, k) \in Q_\beta$, the incidence equation gives

$$u^m w^k = g^m C^k t^{p \cdot (Am - Dk)} = g^m C^k t^{-p \cdot \beta}, \quad (6.3)$$

$$T_\beta(m, k) = t^{p \cdot \beta} \frac{W(Am + \beta, k)}{\prod_a m_a!} u^m w^k. \quad (6.4)$$

Fill the missing indices in $\mathbb{N}^{s+|E|}$ with coefficient zero. Then Corollary 2.3 applies to the entire family in the last expression. Multiplication by the fixed monomial $t^{p \cdot \beta}$ preserves both summability conditions. This proves (v), including the empty sectors.

(v) implies (i). Take $\beta = 0$.

For the last assertion, fix $q_0 \in Q_\beta$. If $q \in S \setminus \{0\}$ has $L(q) \leq 0$, then $q_0 + nq \in Q_\beta$ for every $n \geq 1$. Their leading exponents are $L(q_0) + nL(q)$, and the same descending-support or infinite-multiplicity argument proves failure. The forward implication for this sector was already established in (iv) $\Rightarrow$ (v). $\square$

Corollary 6.2 (A computable obstruction pattern). *If the balancing system fails, there is a finite integer vector $q \in S \setminus \{0\}$ with $L(q) \leq 0$. At least one indecomposable $h \in \mathcal{H}$ occurring in a decomposition of q has $L(h) \leq 0$. Its multiples are an explicit obstruction family.*

Proof. The alternative theorem supplies q . If every indecomposable summand had positive weight, their sum would have positive weight, contrary to $L(q) \leq 0$. $\square$

Corollary 6.3 (Valuation robustness and diagonal invariance). *Keeping the nonzero incidence pattern fixed, replace any coupling or covariance entry by another nonzero Hahn element with the same valuation. Diagramwise summability is unchanged. It is also unchanged by every diagonal monomial transformation (6.2), without requiring that the transformed entries have positive valuation.*

Proof. The criterion depends only on L on S . Equal valuations preserve L . A diagonal transformation changes it by $p \cdot (Am - Dk)$, which is zero on S . $\square$

For real-valued valuation parameters, the admissible set is a finite intersection of homogeneous strict rational half-spaces $L(h) > 0$. For general Γ , the same finite list is interpreted using its order. We will call this the *Wick valuation cone*, with this interpretation understood. Some Hilbert-basis inequalities can be redundant, as the mixed-quartic example will show. A finite generating set of the semigroup and a minimal list of inequalities are different objects.

The case $S = \{0\}$ is not excluded. Its Hilbert basis is empty, the tests are vacuous, and the theorem still produces a balancing vector. The vacuum sum is 1, and every observable sector has finitely many atoms by Proposition 4.3. An individual atom can still have infinite Hahn support.

Divisibility matters for balancing in the original value group. For example, over $\mathbb{C}((t^{\mathbb{Z}}))$ take $P = x^2$ and $C = t$. The vacuum atoms are $(2n-1)!! t^n/n!$, so they are strongly summable. Balancing would require $2p > 0$ and $1 - 2p > 0$, impossible for $p \in \mathbb{Z}$. Passing to rational exponents allows $p = 1/4$, but changes the group in which the certificate is expressed. This example lies outside the standing divisible-group hypothesis and explains why it cannot simply be omitted from condition (iv).

7 Connected diagrams cannot hide the obstruction

The logarithm of a formal Gaussian partition function is classically given by connected diagrams [3, Section 3.5]. It is tempting to expect that restricting to this much smaller family might enlarge the domain of Hahn summability. For interactions without linear vertices, it does not.

A vacuum Wick diagram has m_a vertices of type a . Each vertex has α_i^a distinguishable slots of color i . A pairing produces a multigraph; loops and parallel edges are permitted. The valence of a vertex is $|\alpha^a|$, counting a loop twice. A diagram contributes $g^m C^k / \prod_a m_a!$ when vertices of each type have first been labeled. There are finitely many such pairings for each m . By the connected family we mean the subfamily consisting of the connected pairings, not an unspecified analytic logarithm.

Lemma 7.1 (Indecomposable counts force connectedness). *Every Wick realization of an element $h \in \mathcal{H}$ is connected.*

Proof. Each nonempty connected component has its own nonzero vector of vertex and edge counts in S . If the realization were disconnected, these vectors would decompose h into two or more nonzero elements of S , contradicting indecomposability. Every nonempty component contains a vertex; there are no free-standing edges in a pairing of vertex slots. $\square$

Lemma 7.2 (Cyclic connected amplification). *Let G be a nonempty connected colored Wick graph in which every vertex has valence at least two. If its count vector is h , then for every $n \geq 1$ there is a connected colored Wick graph with count vector nh .*

Proof. The sum of valences is at least twice the number of vertices. Hence the number of edges is at least the number of vertices. A connected multigraph with this property has a non-bridge edge e ; this also includes a loop as a possible choice. Removing e leaves a connected graph.

Take n disjoint labeled copies of G and remove the corresponding edge from each. Let its two exposed slots in copy r be a_r, b_r , of colors i, j , respectively. Re-pair a_r with b_{r+1} , interpreting indices cyclically modulo n . This uses each exposed slot exactly once. Every new edge has the same allowed color pair (i, j) as the removed edge, so all edge-type counts are preserved. Vertex types and slot counts are unchanged. Each punctured copy remains connected and the new edges connect these copies cyclically. The resulting graph is connected and has count vector nh . $\square$

Theorem 7.3 (Equality of connected and full diagramwise domains). *Assume $|\alpha^a| \geq 2$ for every interaction type. The connected vacuum Wick family is strongly summable if and only if the full vacuum family is strongly summable. Equivalently, its exact domain is the finite cone of Theorem 6.1.*

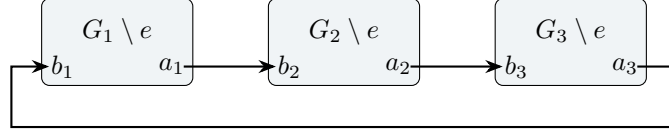


Figure 1: Cyclic amplification for three copies. Each box is connected after one non-bridge edge is removed. Reconnecting the exposed slots preserves every color count. Arrows indicate the re-pairing prescription, not an orientation on Wick edges.

If the criterion fails, there is an indecomposable vector h of nonpositive weight and, for every $n \geq 1$, a connected diagram with leading exponent $nL(h)$. Thus the obstruction can be realized by connected diagrams of arbitrarily large size.

Proof. The connected family is a subfamily of the full family, so full summability implies connected summability. For the converse, if the full criterion fails, Corollary 6.2 gives an $h \in \mathcal{H}$ with $L(h) \leq 0$. The integer incidence equation guarantees that at least one pairing realizes h , by Lemma 3.1. It is connected by Lemma 7.1. Apply Lemma 7.2. The amplitudes of the resulting connected graphs are nonzero rational multiples of $g^{nm}C^{nk}$, so they have leading exponents $nL(h)$. A negative weight produces a descending support sequence; a zero weight produces infinitely many terms with exponent zero. Either contradicts strong summability of the connected family. $\square$

7.1 The connected logarithm on the admissible domain

Proposition 7.4 (Specialization of the connected identity). *Under the equivalent conditions of Theorem 6.1, write $Z = 1 + z$. Then either $z = 0$ or $v(z) > 0$. The Hahn series*

$$\log Z = \sum_{n \geq 1} \frac{(-1)^{n+1}}{n} z^n \quad (7.1)$$

is defined and equals the strongly summed connected vacuum Wick family.

Proof. Every nonempty atom has positive valuation by the criterion. Their family is strongly summable; its support union, when nonempty, is a well-ordered subset of $\Gamma_{>0}$ and therefore has a positive minimum. Cancellation can only raise the valuation of their sum. This proves the assertion about z and makes (7.1) valid by Corollary 2.3.

For completeness, the classical combinatorial identity can be expressed without graph symmetry conventions. Give the vertices of each interaction type separate labels and distinguish their colored slots. Every pairing is uniquely a set of connected pairings on a partition of these labeled vertices. In the exponential generating function, the factors $1/m_a!$ remove the vertex labels; the factors $1/r!$ in an exponential remove the ordering of the r components. Consequently the formal sum of all pairings is the exponential of the formal sum of connected pairings. This is an identity in finitely many formal coupling and edge variables, with only finitely many objects contributing to each coefficient. It is the usual connected-diagram identity [3, Section 3.5].

Use the balancing vector to substitute the positive-valuation entries u_a, w_e of (6.2). Corollary 2.3 respects the formal identity, and the vacuum scaling factors cancel. Taking the formal logarithm, evaluated by (7.1), gives the claimed equality. $\square$

Example 7.5 (Why linear vertices are different). For one variable, $P = gx$ and covariance $C = c \neq 0$, the full vacuum family is

$$Z = \sum_{n \geq 0} \frac{(g^2 c/2)^n}{n!}.$$

It is strongly summable exactly when $v(g^2 c) > 0$. But the connected vacuum family has only the single two-vertex graph and sums to $g^2 c/2$ for every g, c . A graph with all vertices of valence one is a disjoint union of isolated edges, so it has no non-bridge edge to amplify. This shows that the valence hypothesis in Theorem 7.3 cannot simply be deleted. Outside the full domain, the finite connected sum is not being asserted to be the Hahn logarithm of a nonexistent vacuum sum.

8 Positive covariance and sharp stationary-phase scaling

8.1 A collapse to one inequality per monomial

A real symmetric matrix C over $\mathbb{K}_{\mathbb{R}}$ is positive definite if $x^T C x > 0$ for every nonzero $x \in \mathbb{K}_{\mathbb{R}}^d$. This is an ordered-field condition. It is not the statement that every coefficient of every entry is positive.

Lemma 8.1 (Valuation Cauchy–Schwarz). *If C is positive definite, then $C_{ii} > 0$ and, for every nonzero C_{ij} ,*

$$c_{ij} \geq \frac{c_{ii} + c_{jj}}{2}. \quad (8.1)$$

Proof. Positivity on a coordinate vector gives $C_{ii} > 0$. For $i \neq j$, apply positivity to the vector with coordinates $-C_{ij}$ at i and C_{ii} at j . Its quadratic value is $C_{ii}(C_{ii}C_{jj} - C_{ij}^2) > 0$, hence $C_{ii}C_{jj} > C_{ij}^2$. If $2c_{ij} < c_{ii} + c_{jj}$, the negative term $-C_{ij}^2$ would be the leading term of their difference and would make that difference negative. This is impossible. The diagonal case is equality. $\square$

Theorem 8.2 (Positive-covariance criterion). *For positive-definite C over $\mathbb{K}_{\mathbb{R}}$, define*

$$\delta_a = \lambda_a + \frac{1}{2} \sum_i \alpha_i^a c_{ii}. \quad (8.2)$$

The full Wick family is strongly summable if and only if

$$\delta_a > 0 \quad (1 \leq a \leq s). \quad (8.3)$$

The same equivalence holds for an arbitrary diagonal covariance matrix over $\mathbb{K}$ with all diagonal entries nonzero, without a positivity assumption. For positive-definite C , an explicit balancing vector is

$$p_i = \frac{c_{ii}}{2} - \varepsilon, \quad 0 < \varepsilon < \min_a \frac{\delta_a}{|\alpha^a|}, \quad (8.4)$$

whenever (8.3) holds.

Proof. For necessity, fix a and take $m = 2\mathbf{e}_a$. Pair all slots using diagonal edges: set $k_{ii} = \alpha_i^a$ and every cross-edge count to zero. This is a nonzero member of S with weight $2\delta_a$. Theorem 6.1 requires that weight to be positive.

For sufficiency, take any $(m, k) \in S$. By Lemma 8.1,

$$\begin{aligned} \sum_{i \leq j} k_{ij} c_{ij} &\geq \frac{1}{2} \sum_i c_{ii} (Dk)_i = \frac{1}{2} \sum_i c_{ii} (Am)_i, \\ L(m, k) &\geq \sum_a m_a \delta_a. \end{aligned}$$

A nonzero element of S has $m \neq 0$, because $Dk = 0$ with $k \geq 0$ forces $k = 0$. The last expression is therefore strictly positive if all δ_a are positive. Apply Theorem 6.1.

For diagonal covariance, the first inequality is an equality. The same necessity and sufficiency arguments thus apply without ordering its entries. Finally, (8.4) gives

$$\lambda_a + \alpha^a \cdot p = \delta_a - |\alpha^a| \varepsilon > 0,$$

and, by (8.1),

$$c_{ij} - p_i - p_j = c_{ij} - \frac{c_{ii} + c_{jj}}{2} + 2\varepsilon \geq 2\varepsilon > 0.$$

A suitable ε exists by taking half the finite positive minimum in (8.4). $\square$

The theorem concerns a real positive covariance but permits complex couplings. Its diagramwise criterion is insensitive to their phases. Complex phases can nevertheless cause cancellations after grouping; the example in Section 11 uses $C = I$ and does not contradict this theorem because it asks for a different kind of sum.

8.2 An algebraic stationary-phase domain

Consider the polynomial action

$$\mathcal{S}(x) = \frac{1}{2} \sum_{i=1}^d q_i x_i^2 + \sum_{a=1}^s u_a x^{\alpha^a}, \quad q_i, u_a, \hbar \in \mathbb{K}^\times, \quad |\alpha^a| \geq 3. \quad (8.5)$$

Its normalized perturbative Gaussian expression has diagonal covariance $C_{ii} = \hbar/q_i$ and couplings $g_a = -u_a/\hbar$. This is a definition by Wick expansion, not an assertion that an integral over a surcomplex domain has been constructed. It isolates the perturbative part, without an additional Gaussian determinant prefactor.

Corollary 8.3 (Sharp multiscale power counting). *Set $h = v(\hbar)$, $a_i = v(q_i)$, and $b_a = v(u_a)$. The normalized Wick expansion for (8.5) is diagramwise strongly summable exactly when*

$$\Delta_a = b_a + \left(\frac{|\alpha^a|}{2} - 1 \right) h - \frac{1}{2} \sum_i \alpha_i^a a_i > 0 \quad \text{for every } a. \quad (8.6)$$

The same criterion governs every pairable monomial insertion and the connected vacuum expansion. Every polynomial insertion is then defined by the finite linear combination of its monomial sectors; cancellation between those sectors is not needed for the criterion.

Proof. Substitute $\lambda_a = b_a - h$ and $c_{ii} = h - a_i$ into (8.2). This gives $\delta_a = \Delta_a$. The diagonal part of Theorem 8.2 proves the exact criterion, Theorem 6.1 gives the observable statement, and Theorem 7.3 applies because all valences are at least three. $\square$

When the Hessian and interaction coefficients have valuation zero and $h > 0$, the inequalities are automatic for degrees at least three. This is the familiar degree counting behind a formal stationary-phase expansion, not a new discovery; compare [3, Chapters 2–3]. The extra content of (8.6) is its exact necessity and sufficiency when the Hessian and couplings themselves occupy different, possibly non-Archimedean, scales.

For one quartic term, the condition becomes

$$b + h - 2a > 0. \quad (8.7)$$

At equality, infinitely many nonzero diagram atoms have the same leading exponent. Below it, those exponents descend. Thus the strict boundary cannot be repaired by replacing $>$ with $\geq$. A change in the valuation of a degenerating quadratic coefficient can move an otherwise admissible perturbative model across this boundary.

9 Observables, exact identities, and truncation

9.1 Normalized polynomial expectations

Assume henceforth that the main criterion holds. For $f = \sum_{\beta} f_{\beta} x^{\beta} \in \mathbb{K}[x_1, \dots, x_d]$, define

$$\mathcal{Z}_f = \sum_{\beta} f_{\beta} \mathcal{Z}_{\beta}, \quad \langle f \rangle_P = Z^{-1} \mathcal{Z}_f.$$

These are finite sums of well-defined Hahn values. The inverse of $Z = 1 + z$ is given by the strongly summable geometric series $\sum_{n \geq 0} (-z)^n$. In particular, Z is never zero on the diagramwise admissible domain. The construction is linear in f and normalized by $\langle 1 \rangle_P = 1$.

Lemma 9.1 (Algebraic Gaussian integration by parts). *For every polynomial f ,*

$$\mathbb{E}_C[x_i f] = \sum_j C_{ij} \mathbb{E}_C[\partial_j f]. \quad (9.1)$$

Proof. It suffices to take $f = x^{\beta}$. In each pairing of the slots of $x_i x^{\beta}$, the distinguished additional i -slot is paired with one of the β_j slots of color j . Removing that pair leaves a pairing of the slots of $x^{\beta - \mathbf{e}_j}$ and contributes the factor C_{ij} . Summing over its possible partners gives $\sum_{j: \beta_j > 0} \beta_j C_{ij} \mathbb{E}_C[x^{\beta - \mathbf{e}_j}]$, which is the right-hand side. Restricting to $\beta_j > 0$ avoids negative monomial exponents; the omitted polynomial derivatives are zero. $\square$

Theorem 9.2 (Exact perturbative Schwinger–Dyson identities). *For every polynomial f and every i ,*

$$\langle x_i f \rangle_P = \sum_j C_{ij} \langle \partial_j f + f \partial_j P \rangle_P. \quad (9.2)$$

No invertibility assumption on C is needed.

Proof. First treat the coefficients of P as independent formal coupling variables. Apply (9.1) to each polynomial coefficient of $f \exp(P)$ in those variables. Formal differentiation gives

$$\partial_j(f \exp(P)) = (\partial_j f + f \partial_j P) \exp(P).$$

This yields the unnormalized version of (9.2) coefficientwise. Every fixed coefficient involves a polynomial and finitely many pairings.

To justify the specialization, use a balancing vector. Each observable family is the evaluation of a formal power series in the finitely many positive-valuation entries u_a, w_e , multiplied by one fixed monomial as in (6.4). The extra polynomial derivatives introduce only finitely many observable families and fixed Hahn factors. Thus all formal reindexings in the identity are legitimate strong sums. Evaluation gives the unnormalized equality in $\mathbb{K}$. Multiply by Z^{-1} . $\square$

Here ∂_j means ordinary formal differentiation in the polynomial variable x_j . It is not the Berarducci–Mantova derivation on surreal scalars and not an all-scale analytic derivative. The theorem asserts a Hahn-valued algebraic identity, not an integration theorem for a measure.

Formal coupling differentiation similarly gives

$$\frac{\partial Z}{\partial g_a} = \mathcal{Z}_{\alpha^a}, \quad \frac{\partial \log Z}{\partial g_a} = \langle x^{\alpha^a} \rangle_P.$$

These equations are first identities in independent formal coupling variables. Their differentiated series may then be specialized on the admissible domain, by the observable theorem. They do not endow all of $\mathbf{No}[i]$ with a new scalar derivation.

9.2 Support-aware truncation bounds

For a fixed β , let $\mathcal{Z}_{\beta, \leq N}$ be the sum over the finitely many atoms with $|m| \leq N$. Finiteness follows because m has finitely many possibilities and each determines a fixed finite number of slots. Although each individual atom may have infinite Hahn support, this is a finite algebraic expression in the input entries.

Proposition 9.3 (A valuation error estimate). *Choose a balancing vector p and a positive $\eta \in \Gamma$ no greater than the valuations of all scaled entries u_a, w_e . Then*

$$v(\mathcal{Z}_\beta - \mathcal{Z}_{\beta, \leq N}) \geq p \cdot \beta + (N + 1)\eta. \quad (9.3)$$

The convention is that a zero tail has valuation $+\infty$.

Proof. An atom in the tail has $|m| \geq N + 1$. In the scaled expression (6.4), its valuation is at least $p \cdot \beta + (|m| + |k|)\eta$, hence at least the right-hand side of (9.3). Every exponent of the atom is at least its leading exponent. The entire tail is a subfamily of a strongly summable family, and adding its coefficients cannot create an exponent below this common bound. $\square$

The estimate is useful but must not be overinterpreted. In a higher-rank group, the sequence $n\eta$ may be bounded above by a much larger positive element. Thus (9.3) does not generally imply convergence of the partial sums in the intrinsic valuation topology. Strong summability and topological convergence are separate statements.

Example 9.4 (An exact sum with nonconvergent truncations). Take $\Gamma = \mathbb{Q}^2$ with lexicographic order, one variable, $C = 1$, and $P = t^{(0,1)}x^4$. The vacuum sum is

$$Z = \sum_{n \geq 0} \frac{(4n - 1)!!}{n!} t^{(0,n)}.$$

Its support is well ordered, with one nonzero term at each indicated exponent, so it is a Hahn series. The tail after order N has valuation $(0, N + 1)$, which is always less than $(1, 0)$. It never enters the valuation neighborhood requiring error valuation greater than $(1, 0)$. The truncations therefore do not converge to Z in that topology.

Under the explicit embedding in Proposition 2.4, these scales become ω^{-n} and $\omega^{-\omega}$, respectively. The sum is exact as a surreal normal form even though finite truncation does not approximate it at the latter scale.

10 Explicit models and sharp boundaries

10.1 A divergent complex series with an exact surreal value

Let $C = 1$ and $P = -gx^4$. There is one atom per interaction order, and

$$Z(g) = \sum_{n \geq 0} (-1)^n \frac{(4n-1)!!}{n!} g^n = 1 - 3g + \frac{105}{2}g^2 - \frac{3465}{2}g^3 + \dots \quad (10.1)$$

The convention is $(-1)!! = 1$. If a_n denotes the coefficient of g^n , then

$$\frac{a_{n+1}}{a_n} = -\frac{(4n+1)(4n+3)}{n+1}.$$

Its absolute value tends to infinity, so the ordinary complex power series has radius of convergence zero. Nevertheless, Theorem 8.2 says that (10.1) has a strong Hahn value for every g with $v(g) > 0$. For $g = t$ and $t \mapsto \omega^{-1}$ this gives the explicit surreal number

$$1 - 3\omega^{-1} + \frac{105}{2}\omega^{-2} - \frac{3465}{2}\omega^{-3} + \dots$$

The result is a normal-form value, not an assertion that it equals the integral of a Gaussian function over a surreal line.

The connected logarithm begins

$$\log Z(g) = -3g + 48g^2 - 1584g^3 + \dots \quad (10.2)$$

The accompanying program obtains 3, 48, and 1584 independently by enumerating the connected labeled pairings at orders one, two, and three, and compares them with the formal logarithm coefficients. The ordinary combinatorial coefficients are classical; their exact specialization domain is the point of the example.

10.2 An infinite coupling can be admissible

Take two variables and the purely cross covariance

$$C = \begin{pmatrix} 0 & c \\ c & 0 \end{pmatrix}, \quad c \neq 0, \quad P = gx^3 + hy^3.$$

A vacuum pairing requires $3m_1 = k_{12} = 3m_2$, so

$$S = \mathbb{N}(1, 1; 3).$$

There is just one Hilbert generator, and the criterion is

$$v(g) + v(h) + 3v(c) > 0. \quad (10.3)$$

The exact sum is

$$Z = \sum_{n \geq 0} \frac{(3n)!}{(n!)^2} (ghc^3)^n. \quad (10.4)$$

The coefficient counts all bijections between the $3n$ slots of each color.

For $g = t^{-100}$, $h = t^{101}$, and $c = 1$, condition (10.3) is satisfied. One coupling is infinite, yet each realizable vacuum product has increasing valuation. The exact solver returns, for instance,

$$p_x = \frac{67}{2}, \quad p_y = -\frac{403}{12}.$$

The three strict residuals are $1/2$, $1/4$, and $1/12$, respectively. This explicitly demonstrates how an infinite coupling can be compensated by covariance and incidence scaling.

If only the monomial gx^3 is present with the same purely cross covariance, then $S = \{0\}$. Every nonempty vacuum moment vanishes because there are no y -slots to pair with. Thus $Z = 1$ for arbitrary nonzero g , in agreement with the vacuous Hilbert-basis test. Adding hy^3 creates a genuine new cycle condition; checking each interaction in isolation would miss it.

10.3 A mixed quartic instability with a four-element Hilbert basis

Let

$$P = g_1x^4 + g_2y^4, \quad C_{11}, C_{12}, C_{22} \neq 0.$$

Use the coordinate order $(m_1, m_2; k_{11}, k_{12}, k_{22})$. The incidence relations are

$$4m_1 = 2k_{11} + k_{12}, \quad 4m_2 = 2k_{22} + k_{12}.$$

They imply $k_{12} = 2s$ and

$$(m_1, m_2; k_{11}, k_{12}, k_{22}) = (m_1, m_2; 2m_1 - s, 2s, 2m_2 - s), \quad 0 \leq s \leq 2 \min(m_1, m_2).$$

Proposition 10.1 (Exact quartic Hilbert basis and irredundant tests). *The Hilbert basis consists of*

$$\begin{aligned} h_1 &= (1, 0; 2, 0, 0), & h_2 &= (0, 1; 0, 0, 2), \\ h_3 &= (1, 1; 1, 2, 1), & h_4 &= (1, 1; 0, 4, 0). \end{aligned}$$

The summability criterion reduces to the three inequalities

$$\begin{aligned} w_1 &= \lambda_1 + 2c_{11} > 0, \\ w_2 &= \lambda_2 + 2c_{22} > 0, \\ w_4 &= \lambda_1 + \lambda_2 + 4c_{12} > 0. \end{aligned} \quad (10.5)$$

Proof. If $s = 2r$, the count vector is $rh_4 + (m_1 - r)h_1 + (m_2 - r)h_2$. If $s = 2r + 1$, it is $rh_4 + h_3 + (m_1 - r - 1)h_1 + (m_2 - r - 1)h_2$. The indicated coefficients are nonnegative by the bound on s . This proves generation. Each of the four vectors is indecomposable: h_1, h_2 have one vertex, while any nontrivial split of h_3 or h_4 would have to split their vertex vector $(1, 1)$ into $(1, 0) + (0, 1)$, which permits only diagonal pairings and cannot supply their positive cross-edge count.

The weight of h_3 is

$$w_3 = \lambda_1 + \lambda_2 + c_{11} + 2c_{12} + c_{22} = \frac{w_1 + w_2 + w_4}{2}.$$

Thus its inequality follows from the other three. None of those three follows from the other two: with $c_{11} = c_{22} = 0$, one can make either λ_1 or λ_2 negative and choose the other and c_{12} large enough; or take both $\lambda_i > 0$ and choose c_{12} sufficiently negative. This proves the stated reduction and irredundancy. $\square$

Now choose the monomial data

$$C = \begin{pmatrix} t^2 & 1 \\ 1 & t^2 \end{pmatrix}, \quad g_1 = g_2 = t^{-1}. \quad (10.6)$$

Then $w_1 = w_2 = 3 > 0$ and $w_3 = 2 > 0$, but $w_4 = -2$. Each quartic interaction by itself is admissible, while their combination is not. The offending family with $m_1 = m_2 = n$ and all edges cross-colored has atoms

$$\frac{(4n)!}{(n!)^2} t^{-2n}. \quad (10.7)$$

Their leading exponents form a strictly descending sequence. The solver returns precisely $h_4 = (1, 1; 0, 4, 0)$ as its primitive obstruction certificate. The two-vertex four-edge realization is connected, and cyclic amplification gives connected obstructing diagrams at every multiple size.

The matrix in (10.6) is invertible, since its determinant is $t^4 - 1 \neq 0$, but it is not positive definite. Thus this is a legitimate surcomplex bilinear Gaussian model, and it cannot occur under the hypotheses of Theorem 8.2. All its displayed coefficients are positive, so the obstruction is not merely an artifact of a cancellation-sensitive presentation: grouping by interaction order cannot erase it.

10.4 A genuinely higher-rank boundary

In the cross-cubic model, choose $\Gamma = \mathbb{Q}^2$ lexicographically. The entire criterion is the sign of

$$w = v(g) + v(h) + 3v(c) \in \mathbb{Q}^2.$$

A weight $(0, 1)$ is admissible and $(0, -1)$ is not. Both have first coordinate zero. A weight $(1, -10^6)$ is positive and admissible, despite its large negative second coordinate. Replacing the value group by its first real coordinate cannot decide the boundary.

For the inputs $v(g) = (0, 0)$, $v(h) = (0, 1)$, and $v(c) = (0, 0)$, the solver returns

$$p_x = (0, 1/6), \quad p_y = (0, -1/4),$$

with strictly positive residuals $(0, 1/2)$, $(0, 1/4)$, and $(0, 1/12)$. Changing $v(h)$ to $(0, -1)$ instead produces the integer obstruction $(1, 1; 3)$. The same finite incidence problem detects the second-scale transition exactly.

11 Cancellation, grouping, and the precise boundary of the theorem

11.1 An isotropic counterexample to a stronger claim

The criterion is not necessary for every imaginable way of first combining diagrams. This is not a technical concern confined to exotic covariances. Take the positive-definite real covariance $C = I_2$ and the complex linear form $\ell = x + iy$. Its bilinear Wick covariance is

$$\mathbb{E}_C[\ell^2] = 1 + i^2 = 0.$$

By multilinearity of Wick pairings, for every $r \geq 1$,

$$\mathbb{E}_C[\ell^{2r}] = (2r - 1)!!(\mathbb{E}_C[\ell^2])^r = 0. \quad (11.1)$$

Let

$$P = t^{-1}(x + iy)^4. \quad (11.2)$$

After expanding into distinct monomials, every nonzero coupling has valuation -1 , while the diagonal covariance valuations are zero. The exact positive-covariance criterion therefore fails for every interaction monomial. In particular, the $t^{-1}x^4$ monomial alone contributes a subfamily with strictly descending leading exponents.

But grouping the entire polynomial power P^n *before* summing over n gives

$$\frac{1}{n!}\mathbb{E}_C[P^n] = 0 \quad (n \geq 1), \quad Z_{\text{grouped}} = 1. \quad (11.3)$$

Each cancellation in (11.3) is finite, but there are infinitely many such blocks. The raw family is illegal as a Hahn sum while the block family is legal. Finite regrouping preserves an already strong sum; it does not imply the reverse statement when cancellations remove supports from infinitely many blocks.

This example also explains why the criterion is diagonal-monomial invariant but not invariant under arbitrary complex linear changes of coordinates. Using isotropic coordinates can encode a cancellation in the covariance pattern itself. A coordinate-independent classification of all possible cancellation-aware sums is a different problem.

11.2 A class in which order grouping does not help

A Hahn series is *coefficientwise nonnegative* if every coefficient is a nonnegative real number. This is stronger than being positive in the ordered field, which requires only a positive leading coefficient.

Proposition 11.1 (Sign-coherent equivalence). *Suppose every nonzero covariance entry is coefficientwise nonnegative, and $g_a = \sigma u_a$ for a common $\sigma \in \mathbb{C}^\times$, where each u_a is a nonzero coefficientwise nonnegative Hahn series. Then the following are equivalent:*

- (i) *the raw Wick atom family in a fixed sector Q_β is strongly summable;*
- (ii) *the family obtained by grouping all atoms with the same total interaction order $|m| = n$ is strongly summable.*

For a pairable sector, both are equivalent to Theorem 6.1.

Proof. At order n , every atom is σ^n times a nonzero coefficientwise nonnegative Hahn series. Its coefficient multiplier in (3.4) is positive rational. After factoring out the same nonzero complex number σ^n , no coefficient cancels within an order block, and the support of the block is exactly the union of the supports of its atoms. Each block contains finitely many atoms.

If the block family is strongly summable, its support union is well ordered and any fixed exponent occurs in only finitely many blocks. Since every block is finite and no support is erased, the exponent occurs in only finitely many raw atoms. This proves raw strong summability. The other implication is the ordinary regrouping property of strong sums. The last claim follows from Theorem 6.1. $\square$

The proposition includes the common-sign choices $\sigma = \pm 1$ and permits any common nonzero complex factor. It is a sufficient support-preservation hypothesis, not a necessary classification of cancellation-free data.

The mixed obstruction (10.6) satisfies this hypothesis. For the quartic model (10.1), there is already exactly one atom at each interaction order, so raw and order-grouped summability agree for every nonzero Hahn coupling g , even when its coefficients have mixed signs or phases. For example, $g = t - t^2$ has positive valuation but does not become coefficientwise nonnegative after multiplication by any fixed nonzero complex constant. The quartic domain is still exactly $v(g) > 0$. The isotropic example, by contrast, erases supports within its order blocks and shows why the reverse regrouping implication needs a hypothesis.

12 Exact certificate computation and reproducible checks

12.1 What the implementation does

A Python implementation using only the standard library is included in this report's directory as `code/wick_certificates.py`. Its implemented value groups are $\mathbb{Q}^r$ with lexicographic order, for any specified finite $r \geq 1$. A value is represented by a tuple of exact `Fraction` objects. The routine forms the rows A^T and $-D^T$, applies the strict elimination algorithm of Theorem 5.1, and returns one of two certificates: a balancing vector p , or a primitive nonnegative integer obstruction (m, k) . Every returned certificate is rechecked against the original system using exact arithmetic.

For example, with the directory `code` on the Python import path, the mixed-quartic obstruction can be reproduced by

```
from wick_certificates import value, wick_certificate

certificate = wick_certificate(
    alphas=[(4, 0), (0, 4)],
    edges=[(0, 0), (0, 1), (1, 1)],
    coupling_values=[value([-1]), value([-1])],
    covariance_values=[value([2]), value([0]), value([2])],
)
assert certificate.obstruction == (1, 1, 0, 4, 0)
```

The code uses zero-based edge indices, while the mathematical notation uses one-based variable indices. All listed covariance entries are assumed nonzero; omitting an edge is how one specifies a zero entry. The code does not inspect symbolic Hahn series to discover their valuations.

The implementation does not enumerate a general Hilbert basis. It bypasses that potentially expensive task by using the equivalent balancing problem. It also does not construct an arbitrary

Hahn field, decide equality of arbitrary surreal expressions, sum an analytic integral, or implement arbitrary-rank ordered groups. Those would be different computational capabilities. The mathematical theorem allows arbitrary rank; the supplied executable tests use exact finite-rank representations.

12.2 From a failed certificate to connected graphs

There is a finite, though potentially expensive, way to refine an arbitrary integer obstruction q into the indecomposable obstruction used in Theorem 7.3. Search the finite box $0 \leq q' \leq q$ for a nonzero proper solution of the same homogeneous incidence equations. If no such q' exists, q is indecomposable. Otherwise $q = q' + (q - q')$, and at least one summand has nonpositive weight. Replace q by that summand and repeat. The sum of the coordinates decreases strictly, so this terminates.

A pairing with the resulting counts can be built by allocating slots according to the integer edge counts, as in the proof of Lemma 3.1. It is connected automatically. Select a non-bridge edge by checking connectivity after edge removal, and apply the cyclic construction. Thus the obstruction theorem supplies an actual finite recipe for connected counterexamples, not just an existence argument involving a limit or an uncomputable search over all graphs. The supplied program tests the cyclic construction on three types of base multigraph; it does not implement this whole refinement pipeline for arbitrary inputs.

12.3 The recorded test run

Run the checks from the root of a *copy* of this report's directory with

```
python3 code/verify.py
```

The script writes `data/verification.json`. The random test seed is 20260922; no floating-point arithmetic is used in the mathematical checks. The only floating-point quantity in the report is elapsed wall-clock time. One counted case is one call to the script's explicit test-case checker; additional internal assertions verifying every returned certificate are not included in the count.

The output path is fixed relative to the script, not to the working directory, and the file records the elapsed time. Every run therefore rewrites the delivered record, which is why the command above is meant for a copy. The internal certificate rechecks are `assert` statements, so the script must not be run under Python's `-O` flag. The file `code/Makefile` holds optional targets whose paths are relative to the report root; from the root of a copy, `make -f code/Makefile check` runs the suite and `make -f code/Makefile pdf` builds the article. When this report was placed in the collection, the suite was rerun on a copy under Python 3.14.4. It passed all 7,684 cases, and the rewritten record agreed with the delivered one in every entry except the elapsed time.

The feasibility checks include ranks one, two, and three. The moment check uses an independent recurrence obtained by choosing the partner of the first slot, rather than reusing the closed factorial formula. The connected check enumerates all pairings of up to twelve labeled quartic half-edges and tests graph connectivity directly. The cancellation check computes with integers and powers of i , simplified exactly; it is not a numerical near-zero test.

A failed resource guard is reported as an exception. It is not reported as infeasibility, and it does not constitute a mathematical counterexample. This distinction is part of the solver's contract. Concretely, the guard is the `max_rows` argument of the solver, by default 200,000 generated rows,

Finite test family	Cases
Two-quartic semigroup decomposition formula, $0 \leq m_1, m_2 \leq 20$	6,181
Two-quartic feasibility versus its three closed-form inequalities	540
Cross-cubic feasibility versus its single exact inequality	300
Three-variable diagonal feasibility versus the monomial inequalities	210
Vacuous vacuum-semigroup feasibility	75
Colored Wick formula versus an independent moment recurrence	300
Quartic total pairing counts	3
Connected pairing counts versus formal logarithm coefficients	3
Cyclic connected rewiring, including loops and parallel edges	60
Exact isotropic cancellation identities	12
Total passing cases	7,684

Table 1: The recorded finite checks. They test formulas and implementation behavior; the infinite and arbitrary-rank statements are established by the mathematical proofs, not by this table.

and exceeding it raises `RuntimeError`. Fourier–Motzkin elimination can need exponentially many rows, so on large inputs the guard, not the mathematics, may decide whether an answer is returned at all.

13 Novelty audit, sources, and limitations

13.1 Repository scope

The repository inspection was pinned to the main-tree commit

4cf691c7d951e037739d32d9f5c387dcce724f3c

so that its conclusions do not depend on a moving branch. The inspected `docs/README.md` catalogs 26 reports in five families and emphasizes that the research drafts, finite verification programs, and Lean coverage are different kinds of evidence [6]. The catalog includes Hahn analysis, finite deformations, polynomial algebra, spectral theory, dynamics, physics, and several other surreal topics. No report in that catalog is identified as a finite-certificate Wick summability theory.

The opening 190 lines of the physics report were also inspected. They concern the scope of replacing real and complex scalars in physics, including singular spacetime calculations and distinctions between symbolic and physical conclusions. The present work instead fixes a finite algebraic Gaussian model and determines its exact support domain. This is a different question, not a proposed resolution of a spacetime singularity. The source read no further into that report; its passages on perturbative families bear directly on the present subject, and the relation is stated in Section 13.4.

The recursive metadata for `docs/new` listed 18 ZIP archives, including recent work on theta series, Tate uniformization, interpolation, spectral theory, regular singular equations, holonomic rigidity, and dynamics. Their filenames were inspected, but their internal texts were *not* exhaustively read. The repository connector rejected a binary archive fetch, and attempted binary download and snapshot retrieval did not succeed. The source manuscript reported that targeted repository searches for `Wick` and `Gaussian`, made through the connector’s code search, returned no matches, and it treated these as negative search observations that do not prove absence from compressed archives or unindexed content.

Correction made when this report was placed in the collection. For **Gaussian** the reported result is not true of the tree at the pin. A plain text search of the tree at that commit finds the word, in either case, in twenty files; four of them are `article.tex` sources of reports. In the physics report it names a Gaussian regulator: Computation 17.5 there, labelled `phys:comp:gaussian`, whose two ordinary integrals show that an unlimited value defines no product of distributions. Everywhere else it means Gaussian-rational arithmetic (in the computer-algebra, differential-equations, dynamics-and-normal-forms, spectral-theory and infinite-dimensional spectral reports, mostly in their programs and data) or Gaussian elimination (once, in the computer-algebra report). None of these occurrences concerns a Gaussian perturbation expansion, a Wick contraction or a diagram family, so the gap the source identified is real even though its search result was not. For **Wick** the reported result stands: the tree at the pin has no case-sensitive match, and its only case-insensitive match is inside the German word *Entwicklung*, in a bibliography entry of the nonabelian-support report.

Accordingly, the accurate repository claim is: *the theorem package was not found in the repository materials successfully inspected*. A stronger claim that every file and archive in the repository has been checked would be unsupported. The accompanying `RESEARCH_AUDIT.md` records paths, searches, the snapshot identifier, and this access boundary. That file is kept as delivered, so its sentence that both searches returned no matches carries the correction above. The archives have since been opened; see Section 13.5.

13.2 Literature scope and what is not being claimed as new

The following division of credit is part of the mathematical statement of the project.

Ingredient or result	Status in this manuscript
Hahn supports and surreal normal forms	Classical foundations; use [1, 2].
Wick pairing coefficients and the connected logarithm	Classical formal Gaussian combinatorics; compare [3, Chapter 3].
Dickson finiteness and strict elimination	Classical finite-dimensional tools, reproved here; no priority claim.
Exact finite Hahn specialization cone	Proposed contribution in Theorem 6.1, with its observable and robustness conclusions.
Connected obstruction amplification	Proposed contribution in Theorem 7.3; the valence boundary is explicit.
Positive-covariance collapse and multiscale thresholds	Consequences proved in Theorem 8.2 and corollary 8.3, presented as part of the theorem package.
Cancellation and sign-coherent boundary	Explicit limitations and exact comparison results in Section 11.

Table 2: Provenance of the mathematics. “Proposed contribution” means a candidate novelty claim, not certified historical priority.

Etingof’s current author-hosted preliminary book was checked in addition to the arXiv notes; its weighted Feynman theorem and connected-diagram theorem are clear precedents for the formal combinatorics [3]. Neither Wick’s theorem nor formal stationary-phase degree counting is asserted to originate here. Work on higher-rank Hahn geometry, such as Joswig–Smith [4], is also relevant background, but the present argument does not depend on an analytic convergence theorem for multiple indeterminates.

Targeted web searches combining Hahn series, Wick expansions, Gaussian perturbations, valuation balancing, and finite integer certificates did not locate the precise results proved here. Some returned results were only weakly relevant; this makes them particularly unsuitable as evidence of absence. No exhaustive bibliographic search or expert priority review has been completed. The conservative assessment is that the exact combination of results is a plausible new contribution whose novelty remains to be independently checked. This is distinct from uncertainty about what the stated proofs assert: their hypotheses, deductions, and counterexamples are all explicit and available for review.

13.3 Mathematical boundaries

The model is finite-dimensional and has finitely many interaction monomials. Its infinite aspect is the unbounded number of vertices and contractions, not a continuum of field modes. There is no ultraviolet divergence, renormalization construction, path measure, or nonperturbative completion hidden in the notation. The finite criterion does not by itself extend to an infinite list of couplings or covariance entries, where a union of individually good support sets may fail to be well ordered or locally finite.

The theorem is about a prescribed monomial presentation and its Wick pairings. It does not classify all cancellation-aware representations. Strong summability is not ordinary analytic convergence, Borel summability, or convergence of finite truncations in every higher-rank valuation topology. A value in $\mathbf{No}[i]$ is supplied by an explicitly chosen normal-form embedding, with set-sized supports; no unrestricted proper-class integral or sum is introduced.

The exact verification program checks finite algebraic claims. It does not verify the paper in Lean or another proof assistant. Current Lean coverage is recorded separately in `docs/FORMALIZATION.md`: `Surreal/Algebra/WickSemigroup.lean` proves Dickson finiteness, the Hilbert-basis and observable-sector statements, and valuation Cauchy–Schwarz; `Surreal/HahnSeries/NeumannWords.lean` proves the positive-support lemma, including finite word multiplicity. The strict alternative, full Wick summability equivalence, connected-domain theorem and further specialization results are not yet formalized. The build and axiom audit concern the imported declarations listed in that ledger; finite test counts do not extend their proved scope.

13.4 Relation to the physics report

The collection’s physics report, *Surreal Scalars in Theoretical Physics* [7], sorts its own statements into exact symbolic identities, conditional theorems proved in its text, and assessments, and it does not let one kind be read as another. Nothing in the present report belongs to that ledger. Its results are theorems about formal Gaussian coefficient data over a Hahn field. They are not claims about physics, they turn none of that report’s assessments into a theorem, and none of that report’s statements is used here as a premise.

Three passages of that report touch the subject of this one. Section and item numbers below are those of its build when this report was placed; at that time it had not changed since the pin of Section 13.1.

- (a) Its Section 17.2 (`phys:sub:formalpert`) is an assessment. It displays a schematic triple-indexed transseries with algebraic, logarithmic and exponentially small sectors, cautions that an arbitrary such family is not automatically admissible, and adds that a Hahn or surreal

realization does not make an arbitrary formal expansion into a unique actual state or scattering amplitude. Its non-claim N21 (Appendix D.3) repeats the caution.

- (b) Its Computation 17.1 (`phys:comp:factorial`) is an exact identity. The factorial series is a well-defined Hahn element at every positive-valuation parameter, although the ordinary power series has zero radius of convergence, and exact denotation does not give a real-valued sum.
- (c) Its Section 13.5 (`phys:sub:actions`) is an assessment. A proper-class sum over surreal-valued histories is not licensed by Hahn summability, and formal sums, finite-dimensional integrals and functional measures are to be kept apart. Its non-claim N18 repeats the first part.

The relation is the following, and no more.

The physics report poses no open problem on this subject, and this report answers none of its questions. Passage (a) says that admissibility is not automatic. For one precisely delimited class of families, this report decides it exactly. The class consists of the Wick diagram families of a finite polynomial interaction with a symmetric bilinear covariance over $\mathbb{C}((t^\Gamma))$, for divisible Γ : the vacuum family, every pairable observable sector, and, when every valence is at least two, the connected vacuum family. For these, Theorems 6.1 and 7.3 reduce strong summability to finitely many strict valuation inequalities. The mixed quartic model of Section 10.3 is a family of this kind for which strong summability fails although each of its two interactions alone is admissible. This agrees with passage (a) and gives it an exact form for that class. It says nothing about the transseries display of passage (a), whose logarithmic and exponential sectors are not of Wick type, or about any other family.

The quartic series (10.1) is the Wick counterpart of passage (b): an ordinary series of zero radius of convergence has an exact Hahn value at every coupling of positive valuation. The physics report says that exact denotation does not give a real-valued sum. This report says that its value is a normal-form value, not the integral of a Gaussian function over a surreal line (Section 10), and that it asserts no solution to physical divergences (Section 14). The two statements are consistent. Passage (c) is respected throughout: every family summed here is set-indexed in one set-sized Hahn workspace (Section 2), a Gaussian expectation is a finite sum over pairings, and no path measure, functional integral or proper-class sum occurs (Section 13.3). Finally, the isotropic example of Section 11 shows that even inside this class a grouped sum can exist where the diagram family is not summable. Admissibility here is a property of one prescribed family of terms, not of every way of assigning a value to a formal expansion.

13.5 This report in the collection

Provenance. This report is built from a single manuscript, *Finite Certificates for Surcomplex Wick Summability*, dated 22 September 2026 and received as the archive `surcomplex_wick_certificates`, manuscript 12 of the eighteenth batch of external reports placed in the collection. It is not a merge: there was no second source, and nothing was selected out of a larger body of work. Every theorem, proof, example, table and limitation originated in that manuscript; subsequent proof corrections are recorded in the maintained README. Its repository audit is pinned to `4cf691c7d951e037739d32d9f5c387dcce724f3c`. The delivered programs, recorded run and audit notes are kept byte-identical: `code/wick_certificates.py`, `code/verify.py`, `data/verification.json`, `VERIFICATION.md` and `RESEARCH_AUDIT.md`, together with the delivered `Makefile`, which was moved from the package root into `code/`. The delivered README and PDF are not shipped. The PDF is rebuilt from this source.

What changed on placement. Every label now carries the prefix `wick:`, and five subsections received labels they did not have. The title page gained a status note and the abstract a sentence of scope. Section 1.3, Section 13.4 and this subsection are new, and the bibliography gained its entry for the physics report. Section 12.1 now names this report’s directory instead of the delivered archive and says that `code` must be on the import path. Section 12.3 gained the note on running the suite on a copy, the location of the `Makefile`, the `-O` caveat (stated in the delivered `README` and in `VERIFICATION.md`), the size of the resource guard and the record of the rerun. Section 13.1 gained the correction of the `Gaussian` search result and pointers to the two new subsections. Those placement changes did not alter mathematical statements or proofs. The subsequent review recorded in the `README` is separate from placement.

Statements that were true at the pin and are now out of date. At the pin the catalogue `docs/README.md` listed 26 reports in five families; the collection has grown since, and this report is one of the additions. At the pin the 18 archives in `docs/new` could not be opened by the source. They have since been unpacked and their manuscripts placed in the collection’s reports, and the directory `docs/new` no longer exists. When this report was placed, their texts were searched: none contains the word `Wick`, and `Gaussian` occurs in them only as “Gaussian rationals”. In the manuscripts placed with this one, the word means Gaussian rationals, Gaussian elimination or Gaussian quadrature. When this report was placed, no other report in the tree contained `Wick` as a word or any of the terms Feynman, Schwinger, Hilbert basis or Fourier–Motzkin. The source’s provisional conclusion, that the theorem package was not found in the repository materials inspected, therefore extends to a text search of the whole tree at placement. It remains a text search, not a comparison of proofs, and it says nothing about the published literature, where priority is still unchecked (Table 2).

Neighbouring reports. The several-variable multiscale theta series placed in the same batch in the report on Hahn–Tate uniformization is the closest relative. Both characterize strong summability of a monomial family indexed by an integer lattice or semigroup by a finite certificate, both allow Hahn-unit tails and show that such tails cannot repair a failure, and both assume divisible Γ . The mathematics differs: there a quadratic energy on a lattice and its radical flags; here a linear weight on the nonnegative integer kernel S , its Hilbert basis and the strict alternative of Theorem 5.1. Neither report uses a theorem of the other. The distinction between strong summability and convergence of finite subsums, illustrated here by Example 9.4, is drawn throughout the collection, beginning with its foundations report.

14 Conclusions and further questions

The principal result is a complete support criterion for a large but precisely bounded class of surcomplex expressions: finite polynomial Gaussian perturbations with arbitrary symmetric Hahn covariance. Infinite diagram summability is equivalent to a finite strict rescaling problem. Its dual is an integer diagram pattern whose repetitions explain exactly why summability fails. This interpretation survives arbitrary-rank value groups and arbitrary higher Hahn coefficients.

Two refinements give the criterion more than a formal-substitution flavor. The connected sector has the same obstruction domain once linear vertices are excluded, because a non-bridge edge can amplify an indecomposable bad pattern into connected patterns of every size. Positive real covariance, on the other hand, prevents mixed contraction gains and reduces the full cone to one inequality per interaction. These refinements explain both a usable simplification and an explicit mechanism of failure.

The resulting values deserve to be called surcomplex when interpreted through Proposition 2.4: they are actual normal-form numbers, not merely names for unperformed sums. Their interpretation is deliberately algebraic. The quartic example becomes an exact surreal number despite zero complex convergence radius; this says something definite about Hahn support arithmetic without asserting a new measure theory or a solution to physical divergences.

Three natural follow-on questions are newly posed here rather than claimed to be established open problems in the literature. First, can the cancellation-aware domain be described effectively for interactions with specified algebraic symmetries, extending the isotropic example? Second, which additional global support hypotheses permit countably many interaction types while retaining a finite or finitely approximable certificate? Third, for more restrictive graph families such as one-particle-irreducible vacuum graphs, which obstruction patterns admit an amplification preserving that stronger connectivity property? The present proofs identify precisely where each extension would need new input.

A Certificate verification without trusting the solver

A user of the algorithm can ignore its elimination trace and verify the output directly. For a proposed balancing vector p , evaluate the finitely many quantities

$$\lambda_a + \alpha^a \cdot p, \quad c_{ij} - p_i - p_j.$$

Every one must be strictly positive in the specified value-group order. This proves summability by Theorem 6.1.

For a proposed obstruction (m, k) , verify that its coordinates are nonnegative integers, that it is not the zero vector, and that

$$Am = Dk, \quad \lambda \cdot m + c \cdot k \leq 0.$$

These tests prove failure of diagramwise summability without any appeal to the solver. For example, the mixed-quartic witness $(1, 1; 0, 4, 0)$ has

$$A(1, 1)^T = (4, 4)^T = D(0, 4, 0)^T, \quad L = -1 - 1 + 4 \cdot 0 = -2.$$

The cross-cubic positive example has residuals

$$\begin{aligned} -100 + 3 \left(\frac{67}{2} \right) &= \frac{1}{2}, \\ 101 + 3 \left(-\frac{403}{12} \right) &= \frac{1}{4}, \\ 0 - \frac{67}{2} + \frac{403}{12} &= \frac{1}{12}. \end{aligned}$$

All these calculations are rational equalities. In rank two or higher, each equality is evaluated componentwise and positivity is tested lexicographically, using the first nonzero coordinate.

B A dependency map for the proofs

The logical structure can be read without the examples or code. The classical support lemma yields infinitesimal substitution. Integer color balance and the explicit pairing count identify the

exact leading valuation of every atom. Dickson’s lemma supplies a finite Hilbert basis. The strict alternative equates positivity on its integer semigroup with valuation balancing. Necessity follows by repeating one bad pattern; sufficiency follows by balancing and infinitesimal substitution. These are the two directions of Theorem 6.1.

Connected-domain equality additionally uses only the finite-graph facts that an indecomposable count vector has connected realizations and that a connected multigraph of minimum valence two contains a non-bridge edge. The positive-covariance theorem additionally uses only the ordered-field 2×2 positivity inequality. The Schwinger–Dyson identity uses the finite Wick recurrence, followed by the already justified substitution. The counterexample to arbitrary grouping uses the exact algebraic identity $1 + i^2 = 0$.

Thus neither a numerical experiment, an analytic integral, nor an assumed convergence theorem for infinite-rank topologies is a hidden premise of the main proofs. The only imported infinitary algebraic input is the standard Hahn support lemma; the surreal interpretation additionally uses the standard normal-form laws.

References

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- [7] *Surreal Scalars in Theoretical Physics: Black-Hole Singularities, Asymptotic Structure, and the Limits of Replacing the Scalar Field*. Merged research report in the collection, directory docs/physics/surreal-scalars-and-spacetime/. Unchanged between the snapshot of [6] and the placement of this report. Only its Sections 13.5 and 17.2–17.3 and its Appendix D are referred to here, in Section 13.4; none of its statements is used as a premise.